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Short-Term Memory Deficit after Focal Parietal Damage*

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ABSTRACT

The neuropsychological symptomatology is reported for a 44-year-old patient of normal intelligence, EE, after removal of a circumscribed left hemispheric tumor the major part of which was located in the angular gyrus and in the subcortical white matter. EE had a distinct and persistent short-term memory impairment together with an equally severe impairment in transcoding numbers. On the other hand, his performance was flawless in calculation tasks and in all other tests involving number processing. Impairments in language tests could be attributed to his short-term memory deficit, which furthermore was characterized by a strong primacy effect in the absence of a recency effect. His graphomotoric output was temporarily inhibited. The patient, with a strong left-sided dominance, manifested a bi-hemispherical activation of the Broca and Wernicke regions in a positron-emission-tomographic investigation when required to produce verbs which he was to derive from nouns. The findings in EE suggest that unilateral and restricted lateral parietal damage can result in a profound short-term memory deficit together with a transcoding deficit for stimuli extending over only a few digits or syllables in the absence of any symptoms of the Gerstmann syndrome.

Mental processes can be severely and permanently disturbed by small, focal lesions which affect so-called nodal or bottleneck structures of the brain (Cramon & Markowitsch, 1992). Medial diencephalic and medial temporal lobe structures may be seen as examples par excellence for such regions whose bilateral destruction will disrupt the flow of information within complex memory circuits and will consequently result in amnesia (Markowitsch, 1995a; Squire, 1987). Amnesia, defined as a severe long-term memory impairment in patients with otherwise undisturbed intellectual functions (intelligence) and preserved short-term memory abilities, is a common and well-described neuropsychological syndrome. However, in addition to this classic symptomatology, other forms of focal and even unilateral brain damage can result in unique con-

figurations of disturbance which may have major consequences for a person's daily life.

We report the case of a patient with a unique symptom amalgamation which clusters around a short-term memory impairment and a transcoding deficit. The particular case is of interest as a left angular gyrus lesion resulted in neither aphasia nor acalculia or other symptoms of the Gerstmann syndrome (Gerstmann, 1924, 1930; Mazzoni, Pardossi, Cantini, Giorgetti, & Arena, 1990; Shallice & Warrington, 1974; Varney, 1984) yet, the patient had a severe short-term memory impairment and a massive impairment in transcoding numbers, as well as temporary "blocks" in writing and arithmetic tasks.

Damage to the region of the left inferior parietal gyrus (area 40, area PF) may result in agraphia, acalculia, but also in selective agnosia,

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and left-right confusions (Gerstmann syndrome). Sometimes, apraxic disturbances, disturbances in drawing, attention, short-term memory and, more generally, spatial cognition can be found (Critchley, 1953; Thier & Karnath, 1997). The longitudinal fascicle interconnects major portions of the parietal and the prefrontal cortex. Furthermore, shorter fiber bundles from visual and auditory association regions converge in the angular gyrus and provide a basis for multimodal integration in this region.

Short-term memory impairments

In nondemented patients with a severe inability to permanently acquire new information long-term, short-term memory is usually unimpaired (Markowitsch, 1995a; Squire, 1987). The reverse situation of a severe short-term memory impairment and preserved long-term memory is quite a rare event. One of these exceptional cases was patient KF, who was thoroughly studied by Shallice and Warrington (Shallice & Warrington, 1970, 1974; Warrington & Shallice, 1969). Three other patients with this symptomatology were reported by Warrington, Logue, and Pratt (1971) and Saffran and Marin (1975).

Patient KF, a 28-year-old man with a left-hemispheric parietal subdural hematoma, had an auditory verbal retention span of one digit (Warrington & Shallice, 1969), which however did not affect his verbal learning and verbal long-term memory. Two patients with brain damage mainly in the supramarginal and angular gyri of the left hemisphere closely resembled the deficit pattern of KF (Warrington et al., 1971). Saffran and Marin (1975) noted a left posterior parietal lesion for their patient as well, but pointed to the patient's surprising ability in paraphrasing sentences which is indicative of a preserved long-term processing. Shallice and Butterworth (1977) provided convincing evidence that such patients have a specific short-term memory deficit which is not related to the speech output system.

More recently, Butterworth, Cipolotti, and Warrington (1996) described another patient with impaired auditory short-term memory (and preserved calculation skills), and two somewhat less characteristic cases were documented by

Maeshima et al. (1997). These patients had deficits in verbal and nonverbal short-term, but not long-term memory. Both had initially shown some evidence of aphasic disturbances, but although they improved in language functions, their short-term memory deficits remained. The left hemispheres had been traumatically injured in these patients.

Disturbances in number processing

Whereas much earlier reports (Henschen, 1919; Herrmann, 1928; Lewandowsky & Stadelmann, 1908; Singer & Low, 1933) mentioned that damage to the dorsal posterior cortex – or even specifically to the left angular gyrus (Peritz, 1918) – may disturb calculation abilities, it was only recently proven that patients with damage in the region of the angular gyrus may have selective problems with number processing with or without acalculia (Cipolotti, Butterworth, & Denes, 1991; Dehaene & Cohen, 1997; Takayama, Sugishita, Akiguchi, & Kimura, 1994; Warrington, 1982), selective acalculia (Lampl, Eshel, Gilad, & Sarova-Pinhas, 1994), a problem dealing with numbers above a certain, individually differing limit (Cipolotti, Butterworth, & Denes, 1991), a selective processing deficit for certain numbers (Weddell & Davidoff, 1991), or an impairment in reading numerals (Cohen, Dehaene, & Verstichel, 1994). The large symptom variance had already led Berger in 1926 to suggest differentiating between primary and secondary acalculia, with the latter being accompanied by additional impairments in the fields of memory, attention, or language. Another classification with three groups of deficits was proposed by Hécaen, Angelergues, and Houillier (1961): *alexia or agraphia for digits and numbers*, *spatial dyscalculia*, and *anarithmetia* in which the calculation disorder is not related to linguistic or spatial problems.

A more specific impairment, namely, a disturbance in the ability to transcode integers written in digital code (“17”) into the corresponding number in alphabetic script (“seventeen”) or vice versa, has been noted in patients with Broca's or Wernicke's aphasia (Cipolotti et al., 1991; Seron & Deloche, 1983, 1984). More recently, it was also detected in patients with pro-

bable Alzheimer's disease (Klingberg, O'Sullivan, & Roland, 1997; Noel & Seron, 1995; Tegnér & Nybäck, 1990). Although the patients with aphasia made errors by using a term-by-term strategy, numerical displacements, or metalinguistic errors, the demented patients showed further problems not yet observed in other brain-damaged groups, namely, a deficit in the correct use of the two notation systems (e.g. "2 hinderd 11" [2 hundred 11] when trying to transcode the Arabic code "211") (Kessler & Kalbe, 1996).

Finally, Cipolotti and co-workers described two interesting patients, S.A.M. (Cipolotti & Butterworth, 1995) and D.M. (Butterworth & Warrington, 1994) with damage in the parietal region who had preserved or largely preserved calculation abilities, no or only minor aphasic disturbances, but difficulties in transcoding numbers, which were quite selective in one case (Butterworth & Warrington, 1994), and quite pronounced in the other (Cipolotti & Butterworth, 1995). The deficit pattern of our patient EE resembles that of these two cases with respect to his transcoding problems. EE differs from D.M. with respect to his much more severe

and extensive transcoding problems, severe short-term memory impairment, unimpaired calculation ability, strong left handedness and his more circumscribed brain damage. He differs from S.A.M. in his severe short-term memory impairment, the unilaterality and probable etiology of his brain damage, and probably in his handedness (no information given in the report of Cipolotti & Butterworth, 1995).

CASE REPORT

EE, a 44-year-old male left-handed master electrician with his own firm was brought to our attention because of problems with number processing. Pertinent details of the neurological history included the following: Approximately 1½ and 1 years prior to the present examination the patient presented with a grand mal seizure. After the second seizure he underwent cranial computer and magnetic resonance tomography, revealing the likely existence of a tumor in the region of the gyrus angularis in the left posterior cerebral cortex (Fig. 1).

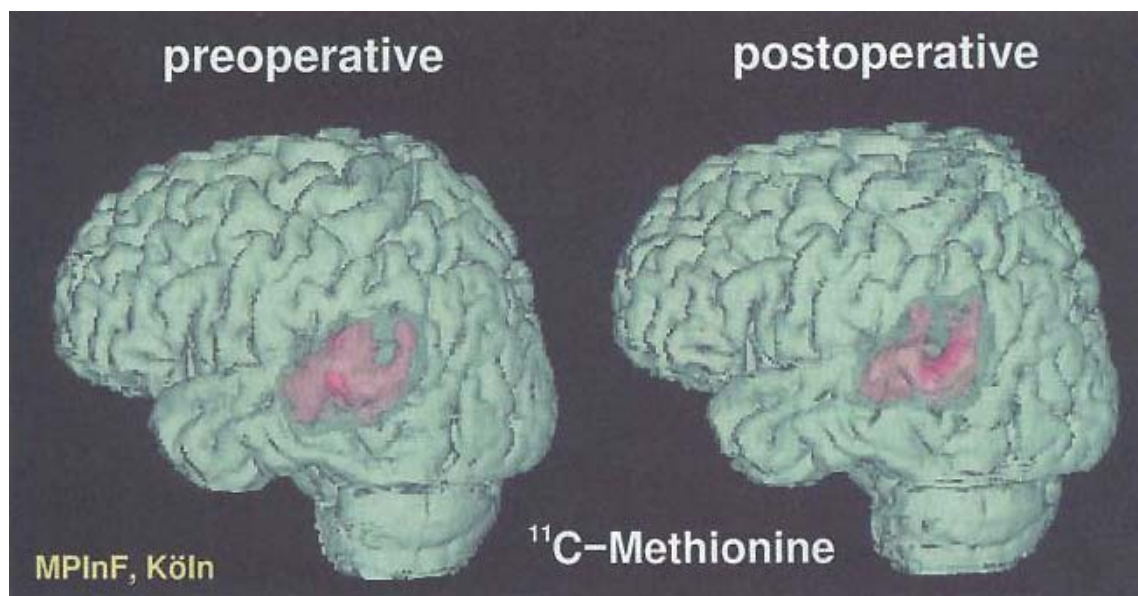


Fig. 1. Three-dimensional reconstruction of brain surface from pre- and post-operative T1-weighted MRI datasets. The area in darker gray in the parietal region represents shape and size of the tumor defined by increased ^{11}C -Methionine uptake measured with PET. The brain tissue around the tumor was made transparent to display the full spatial extent of the tumor beneath the brain surface.

Table 1.

Test	EE	Impairment (= -)	Average or Maximal Score
MMST	29		> 28 ^a
Memo Test, immediate recall	5.2±1.6	—	7.3±0.78 ^b
Memo Test, after delay	7		6.67±1.45 ^b
Rey-Osterrieth Figure, copy	36		33.2±6.1 ^{bc}
Rey-Osterrieth Figure, delayed	15		19.5±6.7 ^c
Reasoning (LPS-IV)	110 (C = 6)		100 (C = 5) ^d
Corsi block span	5		5 ^e
Digit span, forward	3	—	6 ^f
Digit span, backward	3	—	5 ^f
Fluency test, FAS	6+6+6 = 18	—	45.3 ^c
animals	17		18.64 ^c
supermarket items	15	—	28.10 ^g
Modified Trail Making Test	155 s (Percentile = 1)	—	75 s (Percentile = 50) ^h
Drawing (house, flower, clock)	6		6 ⁱ
Stuck-in-set perseveration test	0		0 ^j
Stroop test (NAI)	C = 4.5		C = 5 ^k
T.A.P. , phasic alertness	T = 41		T = 50 ^l
Writing (name, address, sentence)	6		6
Aachener Aphasia Test			
Token Test	10 errors	—	1.25 errors (1.64) ^m
Auditive comprehension	60		53.27 (5.38) ^m
Reading comprehension	58		54.08 (4.27) ^m
Reading aloud	30		29.00 (1.22) ^m
Writing as dictated	24	—	28.49 (1.96) ^m
Naming	118		111.09 (5.38) ^m
Repeating	132	—	146.06 (3.75) ^m
EC 301 R ^m			
Calculating	see text (partial problems)		
Counting	2		2 (0) ^o
Dot enumeration	6		5.7 (0.6) ^o
Placing numbers on an analogue scale	10		9.9 (0.5) ^o
Magnitude judgements, orally presented	8	—	15.5 (0.8) ^o
” ”, written verbal numbers	16		15.4 (0.9) ^o
Perceptual quantity estimation	8		7.9 (0.5) ^o
Contextual magnitude judgements	10		9.8 (1.1) ^o
Mental calculation	13	— (see text)	15.9 (0.5) ^o
Written calculation	11	— (see text)	14.4 (0.9) ^o
Transcoding			
Reading Arabic numbers aloud	32		32 (0) ^o
Reading verbal forms aloud	32		32 (0) ^o
Arabic numbers to written verbal forms	21	—	31.1 (1.6) ^o
Written verbal to Arabic numbers	23	—	31.2 (1.3) ^o
Writing of Arabic numbers from dictation	27	—	31.9 (0.2) ^o
Writing of numerals from dictation	23	—	31.7 (0.6) ^o
Writing telephone numbers transmitted via telephone	18	—	32 (0) ^o
Repetition of numbers	21	—	31.4 (0.3) ^o
Laterality ^p	16		12 = maximal for left; 60 = maximal for right

^aKessler, Herholz, Grond, & Heiss (1991); ^bSchaaf, Kessler, Grond, & Fink (1992); ^cSpreen & Strauss (1991); ^dHorn (1963); ^eSchellig & Hättig (1993); ^fGerman version of the Wechsler Adult Intelligence Test (Tewes, 1991); ^gKessler, Bley, Kerkfeld, Mielke, & Kalbe (1998); ^hOswald & Roth (1987); ⁱGragnaniello, Kessler, Bley, & Mielke (1998); ^jSandson & Albert (1987); ^kOswald & Fleischmann (1995); ^lZimmermann & Fimm (1994); ^mHuber, Poeck, Weniger, & Willmes (1982); ⁿGerman adaptation of the tests of Deloche et al. (1993) by Claros Salinas (EC 301 R) (1994); ^oKalbe (1999); ^pSalmaso & Longoni (1985).

After the second seizure, EE developed mild word finding and number processing problems. As an example, he stated that, when writing down telephone numbers, he made mistakes that he did not recognize until he tried to use the number. No psychiatric data exist from those initial deficits so that their development and course cannot be documented. According to EE's own description, his problems remained rather stable from their first development to the present examinations. The data presented here were collected during four neuropsychological test sessions on 4 consecutive days.

EE's tumor was removed about 5 months prior to the present examinations and was diagnosed histologically as an oligoastrocytoma (Grade II). The patient apparently recovered well from tumor removal. It can be assumed that already during tumor growth a partial functional shift accompanied by the recruitment of especially neighboring and contralateral areas might have taken place (Markowitsch & Calabrese, 1996).

In several neuropsychological test sessions EE showed largely normal performance over a wide range of domains, but severe deficits in short-term memory and in transcoding numbers. EE himself emphasized that these last two impairments had remained unchanged since surgery. Furthermore, he manifested some deficits in language functions, which can, however, be attributed to his auditory short-term memory defect. Calculation and writing were temporarily inhibited. Table 1 gives an overview of the respective results.

EE scored in the above normal range of intelligence (FSIQ = 110; on the German Leistungsprüfsystem IV [LPS IV]; Horn, 1963). He was strongly lateralized to the left. In the laterality questionnaire of Salmaso and Longoni (1985) he responded with 'always left' to 11 of the 12 questions; the only question in which he detailed a preference to the right concerned his eyes. EE's results during a positron-emission-tomographic activation experiment reflect his left-sided lateralization and showed that he activated language regions of the right hemisphere as well as on the left during a verb generation task (Fig. 2).

Language activation study and ^{11}C -methionine PET were performed on a high-resolution PET scanner (Siemens CTI EXACT HR), operating in 3D mode (Wienhard et al., 1994). Images were corrected for randoms, scatter, and measured attenuation. At each activation, 370 MBq [^{15}O]water were administered as a rapid bolus, with 45-s data acquisition time. The activation paradigm comprised three tasks with four replications each in a balanced order: a resting condition (dark room, eyes closed, ears open with low ambient noise), a word repetition task (repeating nouns read aloud) and a verb generation task (finding a semantically matching verb for each noun read aloud). The patient was instructed to repeat nouns or generate words as fast as possible. For each run a new list of nouns was used to avoid recognition and habituation. The words to be repeated were simple high-frequency German nouns. The nouns for verb generation had been selected for high association content (Baschek, Bredenkamp, & Oehrle, 1977).

For the ^{11}C -methionine PET, a bolus of 740 MBq ^{11}C -methionine was injected. Tracer accumulation was recorded in 47 transaxial slices from the entire brain and images were reconstructed from counts accumulated over 20 to 60 min (Herholz et al., 1998b). PET scans were co-registered with each other and with the individual MR image (T1-weighted volume data, 128 slices, 2.5-mm thickness), orientated in parallel to the anterior commissure – posterior commissure line (Pietrzyk et al., 1994). After ratio normalization of each scan to brain average activity, the signal change due to activation was calculated voxel by voxel as a percentage of the reference control value. The result was smoothed by an 8-mm-radius median filter applied only within brain (Herholz et al., 1998a). Fusion images of PET and surface-rendered MRI were generated using a tool described elsewhere (Von Stockhausen et al., 1998).

EE had a somewhat reduced short-term memory for words in a verbal selective reminding task (Memo-Test), but his visuo-spatial short-term memory and both verbal and nonverbal long-term memory were in the average range. His digit span was clearly deficient. In the Logical Memory subtest of the Wechsler Memory

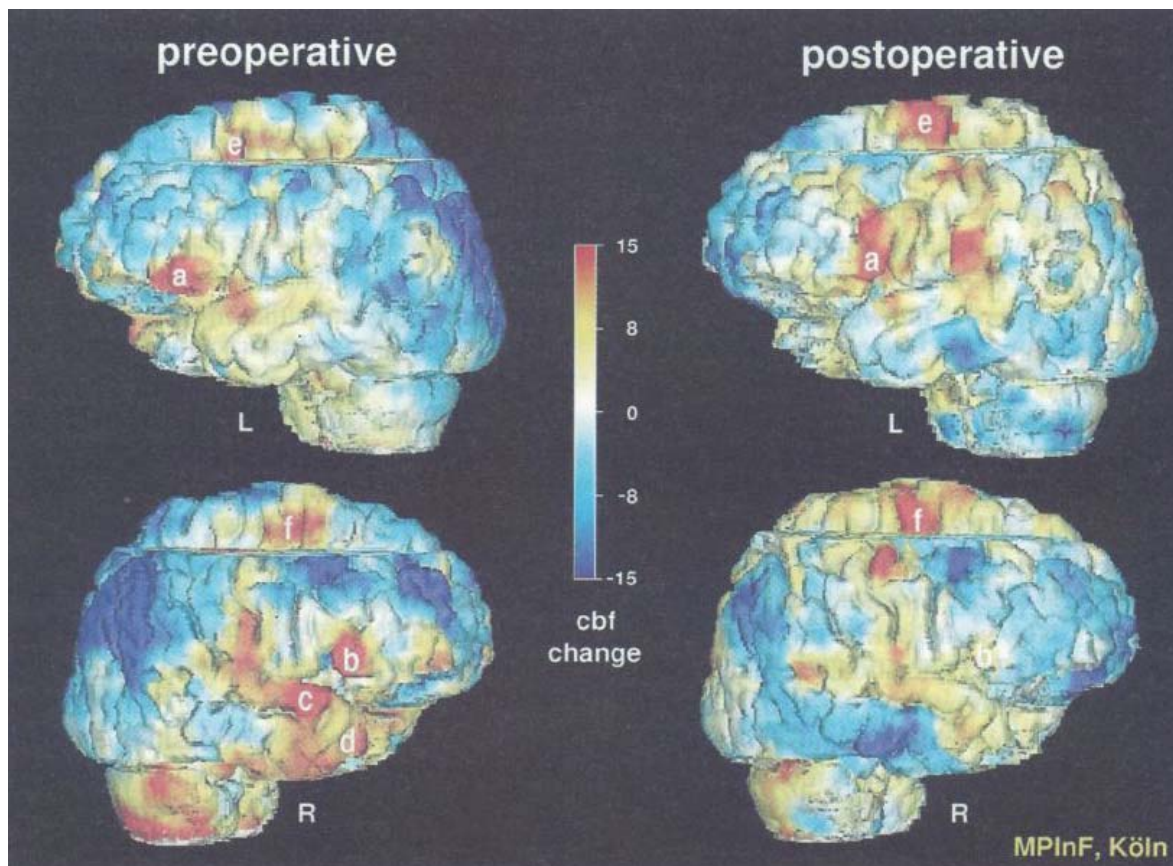


Fig. 2. Three-dimensional reconstructions of brain surface from pre- and post-operative T1-weighted MRI datasets. Changes in cerebral blood flow during a verb generation task (a substantive is given and the patient is asked to respond with the corresponding verb) measured with ^{15}O -water PET are projected onto the brain surface. In order to show the supplementary motor area (SMA) dorsal parts of the contralateral hemispheres were removed in reconstruction. Note the bilateral activation of the Broca region and the higher right-sided activation of the Wernicke area, together with a substantial activation of the auditory cortex (Drury & Van Essen, 1997) and – for the right hemisphere – of surrounding regions of the superior and anterior temporal region, including the temporo-polar cortex. Furthermore, the supplementary motor region is bilaterally activated. (The main cause for the largely reduced activation of the left-sided Wernicke region is, of course, the tumor tissue in this region.)

Scale- revised (Wechsler, 1987), he performed subaverage, but clearly had a superior performance at the 30-min delay re-test as compared to initial story repetition. Furthermore, his processing speed was somewhat reduced. There was no indication for perseverative tendencies and no constructive apraxia. His ability for abstract reasoning was average to slightly above average. Phasic alertness, a measure of arousal according to Pribram and McGuinness (1975), was slightly below average. EE showed no signs

of left-right spatial disorientation or finger agnosia.

In a word generation task EE produced a reduced rate of words. His spontaneous speech was unimpaired. He showed no deterioration in auditory language comprehension and in a reading comprehension task. Errors in the Token Test were indicative of language comprehension problems, but are now considered as attributable to his reduced verbal short-term span (resulting in a quick loss of remembering the instructions).

There were no deficits in reading words or sentences aloud, or in repeating material presented verbally, as long as this material was short. Longer words or sentences were reproduced with errors or omissions, due to his reduced verbal memory span. Thus, for example, instead of 'Farbfernsehgeräteverkauf' he said 'Farbfernsehverkauf' or instead of 'Hallenhandballweltmeisterschaft' 'Hallenball', commenting that he was unable to repeat the whole word. For sentences containing more than six or seven words, he was only able to repeat a few of the first words. Similar problems occurred when repeating numbers. In this task, 11 out of 32 repetitions were incorrect (e.g., the number "seven hundred and six" was repeated as "seven hundred eleven") or could not be remembered at all. In several cases, EE was not conscious of his mistakes. Likewise, his ability for writing words and sentences after a dictation was reduced as he either forgot the words, or had temporary "blocks" (graphematic output disturbance). This impairment is not considered to reflect 'agraphia' in the sense of a systematic linguistic disturbance seen in aphasic patients.

Interestingly, EE had no acalculia and did not show any symptoms of Gerstmann's syndrome. On the EC301R, a German adaptation (Claros, 1994) of a standardized testing battery of calculation and number processing (Deloche & Seron, 1987), he reached full scores in almost all tested subfunctions, namely, counting, dot enumeration, placing numbers on an analogue scale, perceptual quantity estimation, contextual magnitude judgements, and magnitude comparisons of written numbers. Scores in mental arithmetic and magnitude comparison of orally presented numbers were reduced, as EE could not keep the numbers in mind or was blocked during calculation operations. Similar blocks were observed for written arithmetic tasks. When not blocked, EE managed all arithmetic problems flawlessly.

On the other hand, EE manifested severe problems in number transcoding tasks. All applied transcoding tasks (written transcoding from Arabic to verbal numerals and vice versa, writing Arabic and verbal numerals from dictation, and reading Arabic and verbal numerals) consisted

of 32 items, including 8 each of 1-, 2-, 3-, and 4-digit numbers, or a parallel list of number words, depending on the transcoding direction. Larger number sequences were used for the task where telephone numbers had to be written from dictation. Many mistakes occurred in transcoding Arabic numbers into written numerals and vice versa (e.g., to transcode '481' into 'four hundred and eighty one' (11 items wrong) or vice versa (9 items wrong); note that in German multiples of 10 are constructed by inverting the unit and the tens ["four hundred and one and eighty"]). Aside from omissions (e.g., "Fünfhundertsechsfünfzig" (556) → "550"), and inversions (e.g., "86" → "Achtundsechzig" (68)), our patient showed writing blocks, that is, he could temporarily name, but not write numbers or numerals. Such blocks also occurred when trying to write Arabic or verbal numerals from dictation (5 and 11 items wrong, respectively; items had to be repeated in 8 and 17 cases, respectively.) In these tasks, stack position errors (according to the error classification proposed by Deloche and Seron (1987) occurred also, for example, EE wrote "7090" instead of "6090", or "343" instead of "348". It is unclear how many of these errors were "real" lexical errors and how many were caused by his short-term memory deficit. A similar error pattern was performed when writing telephone numbers from dictation. Only reading Arabic and verbal numbers aloud was flawless (only 4 self-corrections in the latter task). In summary, EE's performance in the transcoding tasks were grossly reduced. Part of this impairment is due to his short-term deficit and to temporary "blocks". As transcoding abilities play an important role in everyday life, neuropsychological testing for possible transcoding deficiencies should be included in the routine examination of patients with brain posterior cortical damage.

Thus, EE is a patient with a very selective short-term memory deficit which is apparent especially when trying to operate with numbers and abstract verbal material. Besides that he has a number transcoding deficit. The semantic aspect of numbers is preserved. EE reported that the severity and extent of his selective short-

term memory deficit remained unchanged 6 months after the present examination, leaving him unable to resume work.

DISCUSSION

EE presents as a patient with normal intelligence who has on the average good everyday memory performance, good arithmetic capability, reduced verbal fluency (but no primary aphasic disturbances), reduced attention, a striking verbal short-term memory defect, and a striking deficit in numeral transcoding. Calculation and writing abilities are temporarily inhibited. The cause of this performance pattern seems to lie in a circumscribed left hemispheric parietal lobe damage, caused by a tumor and its removal. EE showed no symptoms of Gerstmann's syndrome such as acalculia, agraphia (at least in the common sense), left-right spatial disorientation, or finger agnosia (Gerstmann, 1930).

Short-term memory processing is not a unitary phenomenon which can be associated with a distinct cortical region. Instead, results from functional imaging studies using both brain-damaged and non-brain-damaged populations indicate that there are a number of – in part sensory-specific – short-term memory components which are embedded in several overlapping cortical networks.

Relating a specific form of short-term memory deficit to unilateral parietal damage is not a new finding (Saffran & Marin, 1975; Vallar, Di Betta, & Silveri, 1997; Warrington et al., 1971; Warrington & Shallice, 1969), though recent data obtained with regional cerebral blood flow measures using positron-emission-tomography (PET) or functional magnetic resonance imaging (fMRI) indicate portions of the prefrontal cortex as the primary target for short-term/working memory processing (Casey et al., 1995; Manoach et al., 1997; Swartz et al., 1996). Past and recent work on animals similarly point to the prefrontal cortex as a short-term memory processor (Crawford, Fulton, Jacobsen, & Wolf, 1948; Jacobsen, 1935, 1936; Jacobsen & Nissen, 1937; Murphy, Arnsten, Goldman-Rakic, & Roth, 1996; Pribram & Tubbs, 1967). Neverthe-

less, there is also some work, for example, obtained with electrophysiological recording techniques, which speaks in favor of a parietal relationship to short-term memory (Constantinidis & Steinmetz, 1996; McCarthy, Luby, Gore, & Goldman-Rakic, 1997), and a recent PET study found bilateral fronto-parietal activation when subjects performed a difficult working memory task (Klingberg, O'Sullivan, & Roland, 1997). Fasotti, Eling, and Bremer (1992), who compared sentence performance on tasks of arithmetic in normal controls, and patients with either frontal or left posterior lesions, found that both frontal and posterior brain damage affects the stage of sentence encoding. Last but not least, there is a strong fiber link between dorsolateral prefrontal and parietal areas (Nieuwenhuys, Voogt, & van Huizen, 1988). The relation of the left-sidedness of the parietal damage to short-term memory impairments can be discussed in light of the HERA-model of Tulving, Kapur, Craik, Moscovitch, and Houle (1994) which predicts that encoding of information is largely a matter of the left hemisphere.

In a PET-activation study Dehaene et al. (1996) found that the inferior parietal gyri were activated especially during multiplication tasks as opposed to tasks comparing numbers, and Whalen, McCloskey, Lesser, and Gordon (1997) reported that stimulation at a left parietal site impaired performance of simple multiplication problems. The data from EE on the other hand show that he can successfully perform multiplication tasks in spite of unilateral inferior parietal damage. However, this comparison should not be overemphasized as our patient was not right-handed and as there is a difference between the regions recruited in an intact versus a damaged neural network as was demonstrated, for example, by Swick and Knight (1996).

The regional cerebral blood flow activations during verb generation (Fig. 2) indicate that EE recruited brain regions which have been implicated in the verbal component of working memory (Paulesu, Frith, & Frackowiak, 1993), namely, in addition to the auditory association regions in the superior temporal gyrus, Broca's area (in fact bilaterally), and also the supramarginal region (especially in the non-damaged right hemi-

sphere). Furthermore and interestingly, EE showed a right-hemispheric activation of the temporo-polar cortex together with a less pronounced activation of the infero-lateral prefrontal region. Both regions have been implicated in long-term memory retrieval both on the basis of PET (Fink et al., 1996; Markowitsch, Thiel, Kessler, von Stockhausen, & Heiss, 1997) and lesion studies (Levine et al., 1998; Markowitsch, 1995b). The bilateral activation of the supplementary motor cortex, observed in EE, was documented as well by Paulesu et al. (1993) and has been related to phonological processing. Fiez et al. (1996) found an activation of the medial portions of the supplementary motor area during short-term maintenance of verbal information.

Of special interest is the discrepancy between EE's preserved calculation and long-term memory capabilities on the one hand, and impaired short-term memory and transcoding abilities on the other. EE resembled the patient of Saffran and Marin (1975) in that he manifested a primacy, but no recency effect when repeating longer words, sentences, or numbers. This behavior, together with his performance in the Memo test (after delay) and in the Logical Memory Test of the WMS-R, is indicative of a preserved long-term memory store, but a deficient "echo box", as Saffran and Marin termed it, or a deficient phonological loop when using Baddeley's (1997) terminology for auditory working memory (Vallar et al., 1997). Again, as in other cases (Butterworth et al., 1996), the conventional working memory concept of Baddeley and Hitch (1974) seems to be at variance with the outcome of preserved calculation ability and an impaired short-term memory, especially for digits. EE's visuo-spatial short-term memory performance on the Corsi task was average. Conduction aphasia as an explanation for EE's deficits in repeating longer sequences of words and sentences can be refuted because of his unimpaired spontaneous speech which showed no signs of the phonematic disturbance usually seen in that type of aphasia.

Several models have been proposed to account for deficits in calculation, number processing, and short-term memory (Cipolotti &

Butterworth, 1995; Dehaene & Akhavein, 1995; Dehaene & Cohen, 1997; Macaruso, McCloskey, & Aliminosa; McCloskey, 1992; McCloskey, Aliminosa, 1993; & Macaruso, 1991; McCloskey, Caramazza, & Basili, 1985; McCloskey, Sokol, & Goodman, 1986; Vallar et al., 1997). Usually several separate but interacting modules are assumed to exist. Vallar et al.'s (1997) anatomico-functional model of the components of phonological short-term memory seems to fit our data quite closely (cf. their Fig. 9). These authors assume that auditory input is phonologically analyzed before entering a phonological short-term store in the inferior parietal lobule. From there it is transmitted to a phonological output buffer in Broca's area and the premotor cortex, finally resulting in spoken output. Rehearsal starts from the last component of this model and returns to the other two. The same rehearsal route may, according to their model, also be used for visual input which is first analyzed and short-term stored, thereafter recoded orthographically and phonologically and then enters the phonological output buffer, resulting either in spoken output or in activation of the rehearsal loop. In line with his brain damage, EE seems to have his greatest impairment in the phonological short-term store. Additional deficits, as seen in his transcoding impairments (i.e., in transposing visually received material) might be explainable by the attachment of visual input to the phonological short-term memory system.

Cipolotti and Butterworth (1995) have modified and expanded the number processing model of McCloskey and co-workers (McCloskey, 1992; McCloskey et al., 1985, 1986, 1991). In the original model, distinct modules for input and output processes for both number words and Arabic numbers are postulated. These are connected to an abstract internal semantic number representation, which has further connections to the calculation system. In this model, number transcoding is only possible via a semantic route. The extended model by Cipolotti and Butterworth (1995) seems to hold especially for the fact that a patient – like EE or their patient S.A.M. – can have intact calculation ability, but an impaired transcoding capability (cf. their Fig.

4) by assuming asemantic routes for number transcoding, as was already suggested by Deloche and Seron (1982). Because EE has no acalculia and shows no deficits in the processing of semantic numeric information (as in magnitude comparison or placing numbers on an analogue scale), but shows severe problems in number transcoding tasks (which can only partly be explained by his short-term memory deficit and temporary blocks), his deficit seems to lie in asemantic transcoding routes as suggested by Cipolotti and Butterworth (1995) and Deloche and Seron (1982) and therefore provides further evidence for their views. In the anatomical and functional model of number processing proposed by Dehaene and Cohen (1995), the transcoding deficit could also be interpreted as a form of disconnection syndrome. Three forms of number representations (an analogue magnitude representation holding for semantic processing, a visual Arabic number form, and a verbal word frame) with different anatomical localizations are connected in a network including diverse parts of the brain of both hemispheres (cf. their Figure 2). A deficit of transcoding Arabic numbers into number words (the first being localized in areas of the left (and partly right) occipitotemporal region and belonging to the ventral visual pathway, the latter being processed "within classical language areas of the left hemisphere, including inferior frontal and superior and middle temporal gyri as well as basal ganglia and thalamic nuclei" (Dehaene & Cohen, 1995, p. 88) and vice versa, could therefore be caused by a disconnection of those areas – in the case of EE caused by a tumor – which themselves remain unimpaired.

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